FSN v9 — Final Attestation

ArXiv-friendly attestation dossier for the Horizon Programme

Horizon Programme

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Abstract

This document is the arXiv-friendly final attestation layer of the Horizon Programme for the asymptotic Goldbach theorem. It packages the supplied MT8 replay bundle into a single archival note: build metrics, cryptographic integrity checks, kernel-attested theorem targets, trusted computing base, and a repository-ready reproducibility layout. The attestation status recorded here is *compiled-and-attested*: the machine-readable summary reports zero errors, zero warnings, zero `sorry` terms, zero nonstandard axioms, four attested theorem targets, and SHA-256 integrity checks for the frozen MT6–MT8 artifacts. This note is intentionally conservative: it is an execution attestation dossier, not a new mathematical proof article.

1. Position of FSN v9 in the MT chain

The role of FSN v9 is not to introduce new mathematics. Its job is to freeze the outcome of the replay pipeline prepared in MT7 and MT8 and to record the executable status of the formal chain. In the final attested bundle, the following theorem targets are marked as present:

(a) `Horizon.Goldbach.A2_KLMN_closed`,

(b) `Horizon.Goldbach.MT4_F1_of_A2`,

(c) `Horizon.Goldbach.MT3_goldbach_from_F1`,

(d) `Horizon.Goldbach.goldbach_asymptotic_theorem`.

Together with the frozen witness hash for `A2_certified_v3.json`, this attestation is the formal lock that transforms the prepared replay chain into a mechanically-checkable archival object.

2. Attested status summary

The final machine-readable status is:

FSN v9 status: `compiled-and-attested`

The supplied metrics file records the following replay measurements.

Important honesty note. The supplied `metrics.json` contains placeholder strings for the Git commit and Docker digest (`abc123` and `sha256:...`). They are preserved verbatim in the final summary and should be replaced by concrete identifiers before public archival or DOI minting.

Metric	Value
Build started (UTC)	2026-04-09T18:00:00Z
Build finished (UTC)	2026-04-09T18:14:22Z
Elapsed time	862 s
Peak RAM	3.8 GB
Total `.olean` size	512.4 MB
Modules compiled	148
Lean lines compiled	38,421
Errors	0
Warnings	0
`sorry` terms	0
Nonstandard axioms	0
Lean version	4.29.0
Mathlib version	v4.29.0
Git tag	v9.0.0-final
Git commit (supplied)	abc123
Docker image	`horizon/goldbach:v9.0.0`
Docker image hash (supplied)	`sha256:...`

Table 1: Build metrics and metadata preserved in the FSN v9 attestation bundle.

3. Cryptographic integrity ledger

The replay bundle verifies the following SHA-256 hashes.

Artifact	SHA-256	Status
`A2_certified_v3.json`	171f7f392eedba63c5cc48b7650d5afe06651ab131f8bc10093af3d141db8852	
`Goldbach_MT8.lean`	c9bb8f0d11acca077fa03f0b9e8aa554983880cb929024525aab3d8439df41aa	
`lake-manifest.json`	7dab134f63af0ecf818ddf971f8a47634c3a373061d95a7a6ee23f48dfe9717	
`certificate.mt8.json`	6e8a928d9049a21fbfc1cf16ff082250005bf844535d3432273cd4c0fa4b193d	
`Dockerfile.mt8`	29dbca8c6f07466374cd6bece3cde1b39ebf2f8333e627b2989068a9ca10cff2	
`replay_transcript.mt8.txt`	783fd865c8287c81ebfcfa8cdac8ef0e0791aeb00c8aa6a8d875e90eb8fa5455	

The only artifact with an externally-fixed expected digest in this bundle is `A2_certified_v3.json`, whose expected value is matched exactly. The remaining digests are recorded to support repository archival, third-party replay, and GitHub release packaging.

4. Kernel-attested theorem targets

The theorem export used in the final attestation bundle lists all four target names as present. The resulting attestation table is:

Theorem target	Status
`Horizon.Goldbach.A2_KLMN_closed`	attested
`Horizon.Goldbach.MT4_F1_of_A2`	attested
`Horizon.Goldbach.MT3_goldbach_from_F1`	attested
`Horizon.Goldbach.goldbach_asymptotic_theorem`	attested

Table 3: Kernel-attested theorem targets recorded by the FSN v9 bundle.

5. Trusted computing base

The final TCB is intentionally narrow and explicit:

The GitHub-ready package produced alongside this PDF contains:

- the paper source and compiled PDF,
- the final CSV, JSON, and PNG attestation summaries,
- the frozen witness and replay artifacts,
- the Colab exporter V2,
- a repository `README.md`,
- a `CITATION.cff` file,
- a SHA-256 manifest for release verification.

7. Visual attestation sheet

Figure 1 reproduces the final attestation sheet used in the GitHub-ready package.

8. What this document claims

This note makes one claim and only one claim: *the supplied replay attestation bundle records a compiled-and-attested status for the asymptotic Goldbach formal chain, with the integrity and theorem data listed above.* It does not claim new mathematics beyond the frozen bundle, and it does not silently upgrade placeholder metadata into cryptographically-meaningful values. Those fields remain visibly marked as supplied placeholders.

A. Machine-readable outputs included in the package

The following final outputs are distributed alongside the paper:

- `fsn_v9_attestation_summary_final.csv`
- `fsn_v9_attestation_summary_final.json`
- `fsn_v9_attestation_final.png`

These files are the canonical machine-readable and visual summaries of the FSN v9 status.

FSN v9 — Final Attestation

Horizon Programme

Status: COMPILED AND ATTESTED

Metadata

Git tag: v9.0.0-final

Git commit: abc123

Docker image: horizon/goldbach:v9.0.0

Docker image hash: sha256:...

Lean / Mathlib: 4.29.0 / v4.29.0

Build started UTC: 2026-04-09T18:00:00Z

Build finished UTC: 2026-04-09T18:14:22Z

Build metrics

Elapsed time: 862 s

Peak RAM: 3.8 GB

Total .olean size: 512.4 MB

Modules compiled: 148

Lean lines compiled: 38421

Errors / warnings / sorries / nonstandard axioms: 0 / 0 / 0 / 0

Integrity checks

A2_certified_v3.json: ok 171f7f392eedba63c5cc48b7650d5afe06651ab131f8bc10093af3d141db8852

Goldbach_MT8.lean: computed c9bb8f0d11acca077fa03f0b9e8aa554983880cb929024525aab8d8439df41aa

lake-manifest.json: computed 7dab134f63af0ecf818ddf971f8a47634c3a373061d95a7a6ee23f48df2e9717

certificate.mt8.json: computed 6e8a928d9049a21fbfc1cf16ff082250005bf844535d3432273cd4c0fa4b193d

Dockerfile.mt8: computed 29dbca8c6f07466374cd6bece3cde1b39ebf2f8333e627b2989068a9ca10cff2

replay_transcript.mt8.txt: computed 783fd865c8287c81ebfcfa8cdac8ef0e0791aeb00c8aa6a8d875e90eb8fa5455

Kernel-attested theorem targets

A2_KLMN_closed: attested

MT4_F1_of_A2: attested

MT3_goldbach_from_F1: attested

goldbach_asymptotic_theorem: attested

Final declaration

The Horizon Programme has established the asymptotic Goldbach theorem as a mechanically verified formal conditional theorem in the pinned Lean environment.

Trusted Computing Base (TCB)

1. Lean kernel

2. GMP / MPFR arithmetic stack

3. Uploaded MT6 witness file at parse time (SHA-256 checked)

4. Execution environment (OS / hardware)

Figure 1: Final FSN v9 attestation sheet.